Hochschule Esslingen
University of Applied Sciences

Fraunhofer
IPA

Master's Thesis

Multi-objective Optimisation of suction-cup and two finger grippers using todo: <Algorithims>

im Studiengang Automotive Systems
der FakultÃ¤t Fahrzeugtechnik
todo: <Semester>

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Eidesstattliche Erklärung

Hiermit versichere ich, die vorliegende Arbeit selbstständig und unter ausschließlicher Verwendung der angegebenen Literatur und Hilfsmittel erstellt zu haben.

Die Arbeit wurde bisher in gleicher oder ähnlicher Form keiner anderen Prüfungsbehörde vorgelegt und auch nicht veröffentlicht.

Esslingen, den June 21, 2026 ________________________

Unterschrift

Sperrvermerk

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Zitat

'Far out in the uncharted backwaters of the unfashionable end of the western spiral arm of the Galaxy lies a small unregarded yellow sun. Orbiting this at a distance of roughly ninety-two million miles is an utterly insignificant little blue green planet whose ape-descended life forms are so amazingly primitive that they still think digital watches are a pretty neat idea.'

Douglas Adams – The Hitchhikers Guide to the Galaxy

Vorwort

Dank an die Firma und die Firmenmitarbeiter, max. 1/2 Seite

Kurz-Zusammenfassung

‘Aushängeschild"’ der Arbeit, max 1 Seite

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Nomenclature

(x_i, y_i)	Planar position of the i-th suction cup
$\mathbf{f}(\mathbf{p})$	Multi-objective function vector
$\mathbf{p}$	Decision variable vector representing the gripper configuration
$\mathcal{G}_{\text{valid}}$	Set of valid (collision-free) grasps
$\mathcal{P}$	Population of candidate solutions
$\mathcal{P}^*$	Set of Pareto-optimal solutions
$\mathcal{P}^{(t)}$	Population at iteration t
$A(\mathbf{p})$	Area of the triangle formed by the three suction cups
$D(\mathbf{p})$	Diversity metric of the grasp distribution
K	Number of clusters (hyperparameter)
N	Population size
t	Iteration index

1 Milestone Timeline

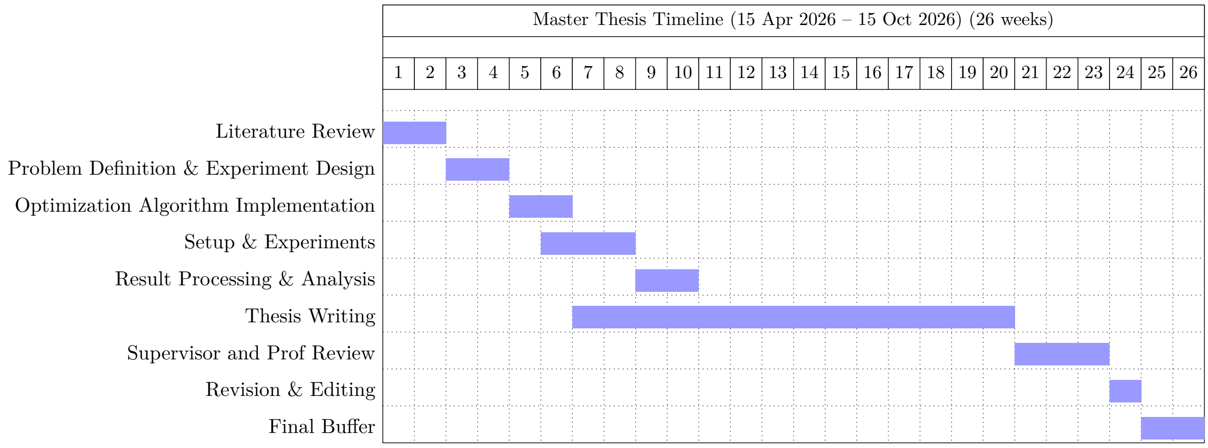

Figure 1.1: Master thesis milestone plan

2 Introduction

2.1 Motivation

Robotic grasping is a fundamental capability in automated manufacturing, logistics, and service robotics. The ability to reliably grasp objects of varying shapes, sizes, and surface properties remains a challenging problem, particularly in unstructured environments. Industrial applications such as bin picking, assembly, and material handling require grippers that can operate robustly across a wide range of workpieces.

Among the commonly used end-effectors, suction-cup grippers and two-finger grippers represent two complementary approaches. Suction grippers are widely used for handling objects with smooth surfaces due to their simplicity and speed, whereas two-finger grippers provide flexibility in handling objects with complex geometries. However, designing optimal configurations for these grippers is non-trivial, as grasp performance depends on multiple interacting factors such as geometry, contact distribution, force limits, and collision constraints.

In particular, multi-suction cup grippers introduce additional complexity due to the spatial arrangement of suction cups. The positioning of the suction cups directly influences grasp stability, force distribution, and robustness to disturbances. Similarly, for two-finger grippers, parameters such as finger geometry and actuator limits affect grasp feasibility and performance.

These challenges motivate the use of systematic optimization techniques to identify high-quality gripper configurations. Since multiple objectives must be considered simultaneously, such as maximizing stability while ensuring robustness and feasibility, the problem is naturally formulated as a multi-objective optimization problem.

2.2 Goal and Tasks of the Thesis

The primary goal of this thesis is to develop and evaluate a multi-objective optimization framework for determining optimal configurations of suction-cup and two-finger grippers.

To achieve this goal, the following tasks are defined:

- Formulation of the gripper configuration problem as a multi-objective optimization problem, including the definition of decision variables, objective functions, and constraints.
- Development of a evaluation pipeline for grasp generation, collision checking, and grasp quality assessment using point cloud representations of workpieces.
- Implementation of an optimization framework capable of handling multiple objectives and generating Pareto-optimal solutions.

- Application and comparison of different optimization algorithms, including Bayesian-based optimization methods, and Evolutionary Algorithms.
- Analysis of the influence of gripper parameters on grasp performance, with particular focus on multi-suction cup arrangements and two-finger gripper configurations.

2.3 Approach

To address the defined objectives, a multi-objective optimization framework is developed. The framework is built upon the *pymoo* library [1], which provides a modular infrastructure for evolutionary multi-objective optimization. A key architectural decision is the decoupling of the optimization logic from the objective function evaluation, which is delegated to an external evaluation service accessed via an API call.

The gripper configurations are parameterized using a set of decision variables provided by the user. Each variable is defined by its type (continuous or integer) and its associated bounds, enabling the framework to handle mixed-variable optimization problems. For the multi-suction cup gripper, the positions of the suction cups are defined in the planar space, forming a triangular arrangement. The height and orientation of the suction cups are assumed to be fixed. Each candidate solution is therefore represented by the coordinates of three suction cups. This work assumes that a triangular arrangement of suction cups leads to a stable grasp configuration [2]. While alternative configurations may exist, the triangular layout provides a minimal and geometrically robust structure for distributing contact forces across the workpiece surface.

An initial population of candidate configurations is generated using mixed-variable sampling and iteratively refined using a user-selected multi-objective evolutionary algorithm. The framework supports multiple algorithms, including NSGA-II, NSGA-III, MOEA/D, CMOPSO, and SMS-EMOA, allowing the user to select the most suitable strategy for the problem at hand.

For each candidate solution, an API call is issued to an external evaluation service, which executes the grasp evaluation subprocess. In this subprocess, grasp candidates are generated for each point of the workpiece point cloud. A collision detection step ensures that only physically feasible grasps are retained by filtering out configurations where the gripper intersects with the workpiece.

Due to the potentially large number of valid grasp candidates, a clustering step is performed to group similar grasps and reduce redundancy. The number of clusters is treated as a hyperparameter. If the number of valid grasps is below a defined threshold, clustering is skipped.

Based on the filtered and clustered grasps, objective function values are computed by the evaluation service and returned to the optimizer. These include the maximization of the area of the triangular arrangement of suction cups, which is associated with grasp stability, and the maximization of a diversity metric that measures the spatial distribution of valid grasp clusters across the workpiece surface. Since *pymoo* operates as a minimization framework, maximization objectives are negated prior to evaluation.

The evaluated objective values are then used by the selected evolutionary algorithm to generate a new population of candidate solutions through selection, crossover, and mutation operators. This iterative process continues until a termination criterion is satisfied. The framework employs a multi-criteria termination strategy: the optimization stops when convergence is detected in either the decision variable space or the objective space (evaluated over a sliding window), or when a maximum number of generations or function evaluations is reached.

Throughout the optimization, the hypervolume indicator is tracked at each generation to monitor convergence toward the Pareto front. Optionally, the framework supports automated hyperparameter tuning of the selected algorithm through a meta-optimization procedure that maximizes the mean hypervolume across multiple random seeds.

The result of the optimization is a set of Pareto-optimal gripper configurations that represent trade-offs between competing objectives, along with the hypervolume progression and total computation time. This information enables the user to select an appropriate configuration based on application requirements.

The same framework is extended to two-finger grippers by adapting the decision variables and objective functions to reflect their specific kinematic and geometric properties. The modular API-based architecture ensures that only the evaluation service needs to be modified, while the optimization pipeline remains unchanged.

3 Methodology

3.1 Methodology for Multi-Suction Cup Gripper Optimization

This section describes the optimization framework developed for a multi-suction cup gripper consisting of three suction cups arranged in a triangular configuration. The framework leverages the multi-objective optimization library *pymoo* [1] and delegates objective function evaluation to an external service via an API call, enabling a modular and decoupled architecture.

3.1.1 Parameterization of the Gripper

Each suction cup is defined by its planar position, while the height and orientation remain fixed. Thus, the decision variables are:

$$\mathbf{p} = [(x_1, y_1), (x_2, y_2), (x_3, y_3)] \tag{3.1}$$

Each individual $\mathbf{p}$ forms a triangular base of the gripper.

Assumption A triangular configuration of suction cups is assumed to yield a stable grasp [2]. This is motivated by the fact that three spatially distributed contact points form a statically stable support, enabling effective load distribution and resistance to external disturbances.

3.1.2 Decision Variables and Bounds

Each decision variable is defined with a type (continuous or integer) and associated bounds:

$$x_i \in [\ell_i, u_i], \quad i = 1, \ldots, n \tag{3.2}$$

The variable definitions are provided by the user as part of the optimization request, allowing the framework to handle mixed-variable problems (i.e., combinations of continuous and integer variables).

3.1.3 Optimization Framework

The optimization is formulated as a multi-objective minimization problem and solved using the *pymoo* library. The user configures the optimization through a request specifying:

- Population size N
- Maximum number of generations G_{max}
- Choice of optimization algorithm
- Number of objectives n_{obj}
- Decision variable definitions (type and bounds)
- Evaluation endpoint URL

Supported Algorithms

The framework supports the following multi-objective evolutionary algorithms:

Table 3.1: Supported multi-objective optimization algorithms.

Identifier	Algorithm
NSGA2	Non-dominated Sorting Genetic Algorithm II [3]
NSGA3	Non-dominated Sorting Genetic Algorithm III [4]
MOEAD	Multi-Objective Evolutionary Algorithm based on Decomposition [5]
CMOPSO	Competitive Multi-Objective Particle Swarm Optimization
SMSEMOA	S-Metric Selection Evolutionary Multi-Objective Algorithm [6]

For algorithms requiring reference directions (NSGA-III, MOEA/D), uniform reference directions are generated using Das and Dennis's method [7]:

$$\Lambda = \texttt{get\_reference\_directions}(\text{“uniform”}, n_{\text{obj}}, n_{\text{partitions}}) \tag{3.3}$$

For mixed-variable problems, specialized sampling and mating operators are employed to handle heterogeneous variable types.

3.1.4 Initialization

An initial population of N candidate solutions is generated via mixed-variable sampling:

$$\mathcal{P}^{(0)} = \{\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_N\} \tag{3.4}$$

3.1.5 Objective Function Evaluation via API

The evaluation of objective function values is decoupled from the optimization loop and delegated to an external evaluation service. For each candidate solution $\mathbf{p}$, the optimizer issues an API call to the evaluation endpoint, which executes the grasp evaluation subprocess (described in Section 3.1.5) and returns the objective values.

$$\mathbf{f}(\mathbf{p}) = \texttt{API\_call}(\mathbf{p}) \rightarrow [f_1(\mathbf{p}),\, f_2(\mathbf{p})] \tag{3.5}$$

This architecture enables:

- Separation of concerns between optimization logic and domain-specific evaluation
- Scalability through independent deployment of the evaluation service
- Reusability of the optimization framework for different problem domains

Grasp Evaluation Subprocess

The evaluation service receives a candidate gripper configuration $\mathbf{p}$ and performs the following steps:

1. **Grasp Generation:** For each point in the workpiece point cloud, a candidate suction grasp is generated by aligning the tool center point (TCP) of the gripper with the surface point.
2. **Collision Filtering:** The gripper is modeled as a rigid 3D body with fixed height, forming a triangular prism whose base is defined by $\mathbf{p}$. Each candidate grasp is checked for collisions between the gripper body and the workpiece. Colliding grasps are discarded.
3. **Clustering of Valid Grasps:** Let $\mathcal{G}_{\text{valid}}$ denote the set of collision-free grasps. If $|\mathcal{G}_{\text{valid}}| \geq K_{\text{min}}$, clustering is performed based on spatial proximity and orientation similarity to reduce redundancy.
4. **Objective Computation:** The service computes and returns the objective values:

$$\mathbf{f}(\mathbf{p}) = [-D(\mathbf{p}),\, -A(\mathbf{p})] \tag{3.6}$$

where:

- $A(\mathbf{p})$ is the triangle area (maximized for stability):

$$A(\mathbf{p}) = \frac{1}{2}\left|(x_2 - x_1)(y_3 - y_1) - (x_3 - x_1)(y_2 - y_1)\right| \tag{3.7}$$

- $D(\mathbf{p})$ is the diversity metric (maximized for robustness), computed from the spatial distribution and orientation spread of clustered grasps.

Since *pymoo* minimizes by convention, maximization objectives are negated.

3.1.6 Optimization Loop

The evolutionary optimization proceeds as follows:

$$\mathcal{P}^{(t+1)} = \mathcal{A}\left(\mathcal{P}^{(t)}, \mathbf{f}\right), \quad t = 0, 1, \dots \tag{3.8}$$

where $\mathcal{A}$ denotes the selected algorithm performing selection, crossover, mutation, and survival operations.

3.1.7 Termination Criteria

The optimization terminates when any of the following conditions is satisfied:

Table 3.2: Termination criteria.

Criterion	Symbol	Description
Design space tolerance	x_{tol}	Change in decision variables below threshold
Objective space tolerance	f_{tol}	Relative improvement in objectives below threshold
Maximum generations	$G_{\max}$	Maximum number of generations reached
Maximum evaluations	$E_{\max}$	Maximum number of function evaluations reached

Tolerances are evaluated over a sliding window of w generations.

3.1.8 Performance Monitoring

The hypervolume indicator [8] is tracked at each generation via a callback mechanism:

$$\mathrm{HV}^{(t)} = \mathrm{HV}(\mathcal{F}^{(t)}, \mathbf{r}) \tag{3.9}$$

where $\mathbf{r} \in \mathbb{R}^{n_{\mathrm{obj}}}$ is the reference point and $\mathcal{F}^{(t)}$ denotes the set of objective vectors at generation t. A monotonically increasing hypervolume indicates convergence toward the Pareto front.

3.1.9 Optional Hyperparameter Optimization

The framework optionally supports automated tuning of algorithm hyperparameters (e.g., crossover probability, mutation rates) using a meta-optimization approach. A `HyperparameterProblem` is constructed around the selected algorithm and evaluated over multiple random seeds. The mean hypervolume across runs serves as the meta-objective:

$$\min_{\boldsymbol{\theta}} \left\{ -\frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \mathrm{HV}(\mathcal{F}_s, \mathbf{r}) \right\} \tag{3.10}$$

The best hyperparameters are then applied to the main optimization run.

3.1.10 Output

The optimization produces a set of Pareto-optimal solutions:

$$\mathcal{P}^* = \{\mathbf{p} \mid \nexists\, \mathbf{p}' \in \mathcal{P} : \mathbf{f}(\mathbf{p}') \preceq \mathbf{f}(\mathbf{p})\} \tag{3.11}$$

The final output includes:

- The Pareto-optimal set $\mathcal{P}^*$ with corresponding objective values
- The hypervolume progression $\{\mathrm{HV}^{(0)}, \mathrm{HV}^{(1)}, \ldots, \mathrm{HV}^{(T)}\}$
- Total optimization wall-clock time

These solutions represent optimal trade-offs between grasp diversity and triangle area (stability).

4 Stand der Technik

Beschreibung des momentanen Standes der Technik und Wissenschaft. Hier muss auch klar definiert werden, in welchem Zusammenhang das Problem steht, d.h. der Systemkontext und der bekannte Lï¿½sungsraum ist klar zu beschreiben (vorhandene.Lï¿½sungen

4.1 Problembeschreibung

4.2 Existierende Lï¿½sungen

4.3 Literaturverweise

Verweise im Text: [?] und [?].

5 Lï¿½sungsansatz

Hier kï¿½nnen je nach Art der Arbeit auch verschiedene Lï¿½sungsansï¿½tze beschrieben und diskutiert werden.

6 Durchgefï¿½hrte Arbeiten

Es muss klar erkennbar sein, auf was aufgesetzt wurde, d.h. schon vorhanden war, und was im Rahmen dieser Arbeit selbst entwickelt wurde.

7 Ergebnisse

Hier kann z.B. eine Bedienanleitung stehen, oder bei Analysen die Analyseergebnisse. 'Neuigkeiten' Messergebnisse

8 Zusammenfassung

Kurze Zusammenfassung und evtl. Beschreibung der Punkte, die zu einer kompletten, marktreifen Lï¿½sung noch fehlen.

9 Schluss

Ergebnis-Bewertung, Zusammenfassung und Ausblick

A Kapitel im Anhang

Alles was den Hauptteil unnötig vergrößert hätte, z. B. HW-/SW-Dokumentationen, Bedienungsanleitungen, Code-Listings, Diagramme

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